

GAIA v3 Command

Reference Manual

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Revision history

Revision	Date	Description
AA	March 2020	Initial release. Alternate document number: CS-00420281-PG
AB	May 2020	Updated for ADK 20.1.
AC	May 2020	Document title changed
AD	August 2020	Added Debug commands
AE	September 2020	Added ANC updates.
AF	December 2020	Updated for ADK 20.3
AG	March 2021	Added: ▪ Get User Feature features ▪ Audio Curation features ▪ Fit Status features
AH	April 2022	Added: ▪ Get Secondary Serial Number command ▪ Secondary Connection Status notification ▪ Handset service feature ▪ Voice processing feature ▪ UI Gesture configuration feature ▪ Statistics reporting feature

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1 GAIA v3 overview

Qualcomm Generic Application Interface Architecture (GAIA) implements an end-to-end, host-agnostic ecosystem supporting host application access to device functionality. This document describes the wire protocol/packet structure of QTIL GAIA and the QTIL GAIA command packet structure and commands. It also gives guidance on porting existing functionality to GAIA v3.

1.1 Differences with earlier releases

- New GAIA Framework domain component
- GAIA Library APIs deprecated
- GAIA over iAP is not supported in this release
- New Vendor ID for QTIL commands
- New packet format of QTIL commands
- GAIA version number is now 3
- Audio Curation Feature ID is now 8

1.2 Porting existing functionality to GAIA v3

This section describes how to port existing GAIA commands to use the GAIA Framework component and the new QTIL command packet format.

Mobile applications

If you are configuring your mobile application needs to support QTIL commands, ensure that you support the following components:

- Support the new QTIL Vendor ID (0x001D)
- Support the new GAIA Version number (3)
- Support the new QTIL command packet structure and protocol

You also need to decide whether you need to support both old and new versions of GAIA.

Devices

Porting existing functionality to GAIA v3 involves:

- Porting functionality to the ADK20.1 application
- Porting existing commands to GAIA v3

To add your commands to GAIA v3, use the GAIA Framework API

```
GaiaFramework_RegisterVendorSpecificHandler().
```

This API registers a callback function that is called when the GAIA Framework receives a Vendor ID that is not the QTIL Vendor ID. This function is passed a `GAIA_UNHANDLED_COMMAND_IND` message.

The GAIA Framework then provides APIs that send responses back to the mobile app.

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2 Protocol and packet reference

2.1 GAIA Wire protocol and packet format

2.1.1 RFCOMM and iAP

Table 2-1 RFCOMM and iAP protocol

0	1	2	3	4+	Length + 1
SOF	Version	Flags	Length	PDU	CS

Where:

SOF is Start of Frame: 0xFF

- Version: 3 or greater
- Flags:
 - Bit 0 set: Checksum in use
 - Bit 1 set: Length extension in use, from wire protocol version 4

Table 2-2 RFCOMM and iAP protocol with bit 1 of the flags set

0	1	2	3	4	5+	Length + 1
SOF	Version	Flags	Length	PDU	CS	

- Bits 1 to 7 are reserved
- Length of Payload:
 - Over 1 byte: Maximum length is 254 bytes
 - Over 2 bytes: the 16-bit length field is big-endian encoded, maximum length is 65534 bytes
- Protocol Data Unit (PDU): See [GAIA PDUs](#).
- CS: Simple XOR of all bytes in the packet. The field is not included when the Bit 0 of the flags is set to 0.

2.1.2 LE GATT

If LE DLE is not present (for example, in models older than iPhone 7), LE GATT packets are much shorter. Therefore LE GATT protocol is simpler to reduce the overhead.

Table 2-3 LE GATT protocol

0+
PDU

Where:

- PDU: See [Gaia PDUs](#).

2.2 GAIA PDUs

Table 2-4 GAIA v3 PDU protocol

Byte 0	Byte 1	Byte 2								Byte 3								Bytes 4+
-	-	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	-
Vendor ID		Feature ID								PDU Type		PDU Specific ID						Payload

Where:

- Vendor ID: Unique, vendor-specific identifier. For example, Bluetooth SIG already has assigned numbers identifying companies.
- Feature ID: Feature-specific unique identifier.
- PDU Type:
 - Command: 00
 - Notification: 01
 - Response: 10
 - Error: 11
- PDU Specific ID: Feature-specific command ID. These are only unique within the Feature.
- Payload: Optional payload.

2.2.1 Command PDU

Direction: Mobile application to the device.

These PDUs are sent by the mobile application to get the device to do something. The PDU-specific ID is called a Command ID.

2.2.2 Notification PDU

Direction: Device to the mobile application.

These are sent by the device to the mobile application generally with status changes. The PDU-specific ID is called a Notification ID.

2.2.3 Response PDU

Direction: Device to mobile application.

This PDU is sent in response to a Command being successfully processed. It may contain a command-specific payload.

2.2.4 Error PDU

Direction: Device to mobile application

This PDU is sent in response to a Command being unsuccessfully processed. The PDU payload contains a status code.

Table 2-5 Default error codes

Status	Code	Description
Failed Feature Not Supported	0x00	An invalid Feature ID was specified.
Failed Command Not Supported	0x01	An invalid PDU Specific ID was specified.
Failed Not Authenticated	0x02	The host is not authenticated to use a Command ID or control.
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Authenticating	0x04	The device is in the process of authenticating the host.
Invalid Parameter	0x05	An invalid parameter was used in the command.
Incorrect State	0x06	The device is not in the correct state to process the command.
In Progress	0x07	The command is in progress.

2.2.5 Status codes

Status codes are 8 bits long where:

- Values 0 to 127 are GAIA Framework values
- Values 128 to 256 are Feature-specific values

2.3 Notifications

Notifications are a GAIA PDU type that enable the device to send information to the mobile application when necessary. To reduce the load on Bluetooth airtime, Feature Notifications are only sent when the mobile application registers to receive them.

The following two commands control Notifications:

- Register
- Unregister

Qualcomm
Confidential - May Contain Trade Secrets
2023-08-03 02:10:03 GMT
hzq@itdtek.com

3 Command reference

3.1 GAIA framework feature

Table 3-1 GAIA framework commands

Feature ID	Version	ADK release
0x00	1	20.1
	2	20.2
	3	21.1
	4	21.1

3.1.1 Get API Version command

The `Get API Version` command gets the GAIA protocol version number, which is 3 or greater.

Table 3-2 Get API Version command

Command ID	Version	ADK release	Variants
0	1+	20.1	All

Command parameters

None

Table 3-3 Get API Version Response PDU contents

Payload byte	Value	Comments
1	≥ 3	GAIA version major
2		GAIA version minor

Possible Feature-specific Error Status codes

None

3.1.2 Get Supported Features command

The `Get Supported Features` command gets the list of features the device supports. If the list is too long to fit into a single packet, then the `More data` field is set, and the `Get Supported Features Next` command should be used to retrieve the remainder of the list.

Table 3-4 Get Supported Features command

Command ID	Version	ADK release	Variants
1	1+	20.1	All

Command parameters

None

Where the `Feature ID Is` bit is aligned into the response packet `Feature ID Is` bit.

If no features are supported, then the Response PDU payload is empty.

Table 3-5 Get Supported Features Response PDU contents

Payload byte	Value	Comments
1	1	More data
1	0	No more data
2	-	1st Feature ID
3	-	1st Feature Version
4	-	2nd Feature ID
5	-	2nd Feature Version
6+	-	List continues

Possible Feature-specific error status codes

None

3.1.3 Get Supported Features Next command

The `Get Supported Features Next` command gets the list continuation of features the device supports.

Table 3-6 Get Supported Features Next command

Command ID	Version	ADK release	Variants
2	2+	20.2	All

Table 3-7 Get Supported Features Next Response PDU contents

Payload byte	Value	Comments
1	1	More data
1	0	No more data

Table 3-7 Get Supported Features Next Response PDU contents (cont.)

Payload byte	Value	Comments
2	-	1st Feature ID
3	-	1st Feature Version
4	-	2nd Feature ID
5	-	2nd Feature Version
6+	-	List continues

Where the `Feature ID 1s` bit is aligned into the response packet `Feature ID 1s` bit.

If no features are supported, then the Response PDU payload is empty.

Possible feature-specific Error Status codes

None

3.1.4 Get Serial Number command

The `Get Serial Number` command gets the customer-provided serial number for this device.

Table 3-8 Get Serial Number command

Command ID	Version	ADK release	Variants
3	1+	20.1	All

Command parameters

None

If no serial number is set, then the Response PDU payload is empty.

Table 3-9 Get Serial Number Response PDU contents

Payload byte	Value	Comments
1+	-	Serial number text

Possible Feature specific Error Status codes

None

3.1.5 Get Variant command

The `Get Variant` command gets the customer-provided variant name.

Table 3-10 Get Variant command

Command ID	Version	ADK release	Variants
4	1+	20.1	All

Command parameters

None

If no variant name is set, then the Response PDU payload is empty.

Table 3-11 Get Variant Response PDU contents

Payload byte	Value	Comments
1+	-	Variant text

Possible Feature-specific Error Status codes

None

3.1.6 Get Application Version command

The `Get Application Version` command gets the customer-provided application version number.

Table 3-12 Get Application Version command

Command ID	Version	ADK release	Variants
5	1+	20.1	All

Command parameters

None

If no application version is set, then the Response PDU payload is empty.

Table 3-13 Get Application Version Response PDU contents

Payload byte	Value	Comments
1+	-	Application version text

Possible Feature specific Error Status codes

None

3.1.7 Device Reset command

The `Device Reset` command enables the mobile app to cause a device to warm reset. The device transmits the response, and then does a warm reset.

Table 3-14 Device Reset command

Command ID	Version	ADK release	Variants
6	1+	20.1	All

Command parameters

None

Response PDU contents

None

Possible feature-specific error status codes

None

3.1.8 Register Notification command

The `Register Notification` command enables the mobile application to register to receive all notifications from a feature.

Table 3-15 Register Notification command

Command ID	Version	ADK release	Variants
7	1+	20.1	All

Table 3-16 Register Notification parameters

Payload byte	Value	Comments
1	-	Feature ID

Response PDU contents

None

Possible Feature specific Error Status codes

None

3.1.9 Unregister Notification command

The `Unregister Notification` command enables the mobile application to unregister to stop receiving feature notifications.

Table 3-17 Unregister Notification command

Command ID	ADK release	Variants
8	20.1	All

Table 3-18 Unregister Notification command parameters

Payload byte	Value	Comments
1	-	Feature ID

Response PDU contents

None

Possible feature-specific error status codes

None

3.1.10 Data Transfer Setup command

The `Data Transfer Setup` command can be used by the mobile application to set up a data transfer channel.

Before using this command, the mobile application needs to send a GAIA Feature command that opens a data transfer session with a Session ID. This 16-bit Session ID and a Transport Type must be specified as `Data Transfer Setup` command parameters.

1. The device checks if both the command parameters (Session ID, Transport Type) are valid.
2. It then finds the feature tied to the Session ID.
3. The device then opens a data transfer channel and binds the channel with the feature's data transfer handler.

Table 3-19 Data Transfer Setup command

Command ID	Version	ADK release	Variants
9	2	20.2	All

Table 3-20 Data Transfer Setup command parameters

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)
3	1	Transport Type: 0: None (Invalid value as the parameter) 1: GAIA_COMMAND_LINK

Table 3-21 Data Transfer Setup Response PDU contents

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)

Table 3-22 Data Transfer Setup command possible error status codes

Status	Code	Description
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Invalid Parameter	0x05	An invalid parameter was used in the command.
Incorrect State	0x06	The device is not in the correct state to process the command.

3.1.11 Data Transfer Get command

The `Data Transfer Get` command can be used by the mobile application to request the device to send data bytes as the response.

Before using this command, the mobile application must set up a data transfer channel to the device with the `Data Transfer Setup` command.

Table 3-23 Data Transfer Get command

Command ID	Version	ADK release	Variants
10	2	20.2	All

Table 3-24 Data Transfer Get command parameters

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)
3	(any)	Start offset (MSB)
4	(any)	Start offset (Second SB)
5	(any)	Start offset (Third SB)
6	(any)	Start offset (LSB)
7	(any)	Data size (MSB)
8	(any)	Data size (Second SB)
9	(any)	Data size (Third SB)
10	(any)	Data size (LSB)

Table 3-25 Data Transfer Get Response PDU contents

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)
3	(any)	Data (Offset + 0)
4	(any)	Data (Offset + 1)
5	(any)	Data (Offset + 2)
...	...	...
$N + 3$	(any)	Data (Offset + N)

Table 3-26 Data Transfer Get command possible error status codes

Status	Code	Description
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Invalid Parameter	0x05	An invalid parameter was used in the command.
Incorrect State	0x06	The device is not in the correct state to process the command.

NOTE GAIA Debug Feature uses `Data Transfer Get` command to read panic log from the device.

3.1.12 Data Transfer Set command

The `Data Transfer Set` command can be used by the mobile application to send data bytes to the device.

Before using this command, the mobile application must set up a data transfer channel to the device with the `Data Transfer Setup` command.

Table 3-27 Data Transfer Set command

Command ID	Version	ADK release	Variants
11	2	20.2	All

Table 3-28 Data Transfer Set command parameters

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)
3	(any)	Start offset (MSB)
4	(any)	Start offset (Second SB)
5	(any)	Start offset (Third SB)
6	(any)	Start offset (LSB)
7	(any)	Data (Offset + 0)
8	(any)	Data (Offset + 1)
9	(any)	Data (Offset + 2)
...	...	...
$N + 7$	(any)	Data (Offset + N)

Table 3-29 Data Transfer Set Response PDU contents

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)

Table 3-30 Data Transfer Set command possible error status codes

Status	Code	Description
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Invalid Parameter	0x05	An invalid parameter was used in the command.
Incorrect State	0x06	The device is not in the correct state to process the command.

3.1.13 Get Transport Information command

The `Get Transport Information` command can be used by the mobile application to get information about the GAIA transport.

Table 3-31 Get Transport Information command

Command ID	Version	ADK release	Variants
12	2	20.2	All

Table 3-32 Get Transport Information command parameters

Payload byte	Value	Comments
1	-	Transport key: ■ 1: Maximum packet size in bytes the device can send ■ 2: Optimum packet size in bytes the device should send. Optimum is generally taken to mean fastest transfer. ■ 3: Maximum packet size in bytes the device can receive. ■ 4: Optimum packet size in bytes the device should receive. Optimum is generally taken to mean fastest transfer. ■ 5: Transport flow control in device to phone direction. □ 1: Flow control □ 0: No flow control ■ 6: Transport flow control in phone to device direction. □ 1: Flow control □ 0: No flow control ■ 7: Current protocol version

NOTE For RFCOMM and iAP2, `PACKET_SIZES` are the size of the GAIA packet and do not include any transport or lower protocol headers. For Bluetooth Low Energy/GATT, the `PACKET_SIZES` include the 3 byte ATT header, so a reported size of 65 actually means a GAIA packet size of 62.

Table 3-33 Get Transport Information response PDU contents

Payload byte	Value	Comments
1	-	Transport key from command
2	(any)	Value of requested key as unsigned 32-bit value, in big endian format
3	(any)	
4	(any)	
5	(any)	

Table 3-34 Get Transport Information command possible error status codes

Status	Code	Description
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Invalid Parameter	0x05	An invalid parameter was used in the command.

3.1.14 Set Transport Parameter command

The `Set Transport Parameter` command can be used by the mobile application to request changing a GAIA transport parameter. If the parameter value is accepted it will be returned in the response, if the value is not accepted the old value or a value that is as close as possible to the requested value will be returned.

Table 3-35 Set Transport Parameter command

Command ID	Version	ADK release	Variants
13	2	20.2	All

Table 3-36 Set Transport Parameter command parameters

Payload byte	Value	Comments
1	-	Transport key: 1: Maximum packet size in bytes the device can send 2: Optimum packet size in bytes the device should send. Optimum is generally taken to mean fastest transfer. 5: TX_FLOW_CONTROL 7: Protocol version
2	(any)	Requested value of key as unsigned 32-bit value, in big endian format
3	(any)	
4	(any)	
5	(any)	

Table 3-37 Set Transport Parameter response PDU contents

Payload byte	Value	Comments
1	-	Transport key from command
2	(any)	Actual value of key as unsigned 32-bit value, in big endian format
3	(any)	
4	(any)	
5	(any)	

Table 3-38 Set Transport Parameter command possible error status codes

Status	Code	Description
Failed Insufficient Resources	0x03	The command was valid, but the device could not successfully carry out the command.
Invalid Parameter	0x05	An invalid parameter was used in the command.

3.1.15 Get User Feature command

The `Get User Feature` command can be used by the mobile application to read the user feature data from the device. If the data is available, the data starting from the first byte is set to the response payload. If the whole data does not fit into the response payload, the `MoreData` bit in the response is set to 1. The remaining data can be read with the `Get User Feature Next` command.

Table 3-39 Get User Feature command

Command ID	Version	ADK release	Variants
14	3	21.1	All

Command parameters

None.

Table 3-40 Get User Feature response PDU contents

Payload byte	Value	Comments
1	0x00 or 0x01	- Bits [7:1]: Reserved for future use. - Bit [0]: <code>MoreData</code>. If this bit is set to 1, there is remaining data that did not fit in this response payload.
2	(any)	Reading Status: This three-byte status parameter is used as the parameter for the <code>Get User Feature Next</code> command to read the remaining data.
3		
4		
5	(any)	Data bytes of the user feature data (if any).

Table 3-41 Get User Feature command possible error status codes

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command.

3.1.16 Get User Feature Next command

The `Get User Feature Next` command can be used by the mobile application to read the continuous part of the user feature data, which has not been read by the last `Get User Feature (Next)` command.

If the data is available, the continuous data is set to the response payload. If the remaining data does not fit into the response payload yet, the `MoreData` bit in the response is set to 1. Another `Get User`

Feature Next command or commands can be used to read the following data until the mobile application receives the response with the MoreData bit cleared to zero.

Table 3-42 Get User Feature Next command

Command ID	Version	ADK release	Variants
15	3	21.1	All

Table 3-43 Get User Feature Next command parameters

Payload byte	Value	Comments
1	(any)	Reading Status: The three-byte status parameter that has been returned from the device as a part of the response of the preceding Get User Feature (Next) command.
2		
3		

Table 3-44 Get User Feature Next command response PDU contents

Payload byte	Value	Comments
1	0x00 or 0x01	- Bits [7:1]: Reserved for future use. - Bit [0]: MoreData. If this bit is set to 1, there is remaining data that did not fit in this response payload.
2	(any)	Reading Status: This three-byte status parameter is used as the parameter for the Get User Feature Next command to read the remaining data.
3		
4		
5	(any)	Data bytes of the user feature data (if any).

Table 3-45 Get User Feature Next command possible error status codes

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command.

3.1.17 Get Device Bluetooth Address command

The Get Device Bluetooth Address command gets the permanent (BR/EDR) Bluetooth Device address. For a pair of earbuds this is the address of the primary earbud.

Table 3-46 Get Device Bluetooth Address command

Command ID	Version	ADK release	Variants
16	4		All

Command parameters

None.

Table 3-47 Get Device Bluetooth Address Response PDU contents

Payload byte	Value	Comments
1	-	NAP, MSB
2	-	NAP, LSB
3	-	UAP
4	-	LAP, MSB
5	-	LAP
6	-	LAP, LSB

Possible Feature specific Error Status codes

None.

3.1.18 Charger Status notification

The Charger Status notification enables the device to generate a notification when the charger is plugged in or unplugged.

Table 3-48 Charger Status notification

Notification ID	Version	ADK release	Variants
0	1	20.1	All

Table 3-49 Charger Status notification parameters

Payload byte	Value	Comments
1	0x00	Unplugged
	0x01	Plugged in

3.2 Earbud Application feature

Table 3-50 Earbud Application commands

Feature ID	Version	ADK release
0x01	1	20.1
	1	20.2
	2	20.3

3.2.1 Which Earbud is Primary command

The `Which Earbud is Primary` command returns whether the primary is the Left or Right earbud. This can be used for example to match the serial number to the Left or Right earbud.

Table 3-51 Which Earbud is Primary command

Command ID	Feature version	ADK release	Variants
0	1+	20.2	All

Command parameters

None

Table 3-52 Which Earbud is Primary command parameters

Payload byte	Value	Comments
1	0 or 1	■ 0: Left ■ 1: Right

Possible feature-specific Error Status codes

None

3.2.2 Get Secondary Serial Number command

The `Get Secondary Serial Number` command gets the customer-provided serial number for the secondary device.

Table 3-53 Get Secondary Serial Number command

Command ID	Feature version	ADK release	Variants
1	2+	21.1	All

Command Parameters

None

Table 3-54 Get Secondary Serial Number Response PDU parameters

Payload byte	Value	Comments
1+	-	Serial number text

If no serial number is set, then the Response PDU payload is empty.

Possible feature-specific Error Status codes

None

3.2.3 Primary Earbud About To Change notification

The device can generate a notification when the primary device changes. This happens due to handover where the secondary earbuds have a better link to the mobile phone.

Table 3-55 Primary Earbud About To Change notification

Notification ID	Feature version	ADK release	Variants
0	1+	20.1	All

Table 3-56 Primary Earbud About to Change notification parameters

Payload byte	Value	Comments
1	0 or 1	0: Static handover of the GAIA link that is, it is disconnected, and the mobile app needs to reconnect it after waiting for five seconds. No state is retained in the device. 1: Dynamic handover of the GAIA link that is, it remains connected during handover but notifications need to be re-registered.
2	0 to 255	Time to delay any re-connection attempt to allow handover to complete. Value is in seconds.

3.2.4 Primary Earbud Changed notification

Primary Earbud Changed notification is sent by the primary earbud after the earbuds have changed roles.

Table 3-57 Primary Earbud Changed notification

Command ID	Feature version	ADK release	Variants
1	2	20.3	All

Table 3-58 Primary Earbud Changed notification parameters

Payload Byte	Value	Comments
1	0 or 1	■ 1: Left ■ 0: Right

3.2.5 Secondary Earbud Connection State notification

Secondary Earbud Connection State notification is sent by the primary earbud when it connects to or disconnects from the secondary earbud.

Table 3-59 Secondary Earbud Connection State notification

Notification ID	Feature version	ADK release	Variants
2	3	21.2	All

Table 3-60 Primary Earbud Changed notification parameters

Payload byte	Value	Comments
1	0 or 1	■ 1: Connected ■ 0: Disconnected

3.3 Voice UI (VA) feature

Table 3-61 VA commands

Feature ID	Version	ADK release
0x03	1	20.1

3.3.1 Get Selected Assistant command

The `Get Selected Assistant` command returns the currently active Voice Assistant.

Table 3-62 Get Selected Assistant command

Command ID	Feature version	ADK release	Variants
0	1+	20.1	All

Command parameters

None

Table 3-63 Get Selected Assistant Response PDU Contents

Payload byte	Value	Comments
1	-	Identifier of currently selected assistant

Assistant identifiers in the range 0x00 to 0x7F are assigned by QTIL. IDs in the range 0x80 to 0xFF are for customers' own use.

Table 3-64 Assistant identifiers

ID	Assistant
0x00	None
0x01	Reserved
0x02	GAA (Google)
0x03	AMA (Alexa)

3.3.2 Set Selected Assistant command

The `Set Selected Assistant` command selects the active Voice Assistant.

Table 3-65 Set Selected Assistant command

Command ID	Feature version	ADK release	Variants
1	1+	20.1	All

Table 3-66 Set Selected Assistant parameters

Payload byte	Value	Comments
1	-	Identifier of assistant to be selected

Response PDU Contents

None

3.3.3 Get Supported Assistants command

The `Get Supported Assistants` command returns a list of supported Voice Assistants. The list will include `0x00` ('none') if the device supports switching to no assistant.

Table 3-67 Get Supported Assistants command

Command ID	Feature version	ADK release	Variants
2	1+	20.1	All

Command parameters

None

Table 3-68 Get Supported Assistants Response PDU Contents

Payload byte	Value	Comments
1	n	Number of supported assistants
2	-	Identifier of first assistant
...	...	...
n + 1	-	Identifier of n th assistant

3.3.4 Assistant Changed notification

Assistant Changed notification is sent when the active Voice Assistant is changed.

Table 3-69 Assistant Changed notification

Notification ID	Feature version	ADK release	Variants
0	1+	20.1	All

Table 3-70 Assistant Changed notification parameters

Payload byte	Value	Comments
1	-	Identifier of currently selected assistant.

3.4 Debug commands

Table 3-71 Debug commands

Feature ID	Version	ADK release
0x04	1	20.2
	2	20.3

Debug commands provide two different debugging features:

- Reading debug information saved at a panic
- PyDbg access to the device

3.4.1 Get Debug Log Info command

The `Get Debug Log Info` command returns the information about the debug partition and panic log stored in it.

Table 3-72 Get Debug Log Info command

Command ID	ADK release	Variants
0x01	20.2	All

Table 3-73 Get Debug Log Info command parameters

Payload byte	Value	Comments
1	(any)	Debug Log Info Key (MSB)
2	(any)	Debug Log Info Key (LSB)

Table 3-74 Debug Log Info Keys

Value	Comments
0x0000	Debug partition size
0x0001	Panic log size

Table 3-75 Get Debug Log Info Response PDU contents

Payload byte	Value	Comments
1	0	Debug Log Info Response PDU Version
2	(any)	Debug Log Info Key (MSB)

Table 3-75 Get Debug Log Info Response PDU contents (cont.)

Payload byte	Value	Comments
3	(any)	Debug Log Info Key (LSB)
4	(any)	Size of 'debug partition' or 'panic log' (MSB)
5	(any)	Size of 'debug partition' or 'panic log' (Second SB)
6	(any)	Size of 'debug partition' or 'panic log' (Third SB)
7	(any)	Size of 'debug partition' or 'panic log' (LSB)

Table 3-76 Get Debug Log Info command possible error status codes

Status	Code	Description
Invalid parameters	0x85	Invalid Debug Log Info Key.
No debug partition	0x86	The debug partition is not found. (The partition must be defined in the flash layout config file.)
Failure by unknown reason	0x8F	Unspecified reason caused the failure of the requested operation.

3.4.2 Configure Debug Logging command

The `Configure Debug Logging` command sets a `debug_partition` config parameter, with which the developer can configure the items of the debug information to be saved at a panic (see *Using the Debug Partition Application Note*).

This command passes the Config Key and Config Value to the `debug_partition` configuration function on the device, and it receives the result.

Table 3-77 Get Debug Logging command

Command ID	ADK release	Variants
0x02	20.2	All

Table 3-78 Configure Debug Logging command parameters

Payload byte	Value	Comments
1	(any)	Debug Log Config Key (MSB)
2	(any)	Debug Log Config Key (LSB)
3	(any)	Debug Log Config Value (MSB)
4	(any)	Debug Log Config Value (Second SB)
5	(any)	Debug Log Config Value (Third SB)
6	(any)	Debug Log Config Value (LSB)

Table 3-79 Configure Debug Logging Response PDU contents

Payload byte	Value	Comments
1	(any)	Debug Log Config Key (MSB)
2	(any)	Debug Log Config Key (LSB)
3	(any)	Debug Log Config Value (MSB)
4	(any)	Debug Log Config Value (Second SB)
5	(any)	Debug Log Config Value (Third SB)
6	(any)	Debug Log Config Value (LSB)

Table 3-80 Get Debug Log Info command possible error status codes

Status	Code	Description
Not enough space	0x82	No space is available to store panic log.
Invalid parameters	0x85	Invalid Debug Log Config Key.
No debug partition	0x86	The debug partition is not found. (The partition must be defined in the flash layout config file.)

3.4.3 Get Panic Log command

The `Get Panic Log` command initiate the transfer of the panic log from the device. This command must be followed by `Data Transfer Setup` command to set up the data transfer path, and then `Data Transfer Get` command to read panic log from the device. See sections [Data Transfer Setup command](#) and [Data Transfer Get command](#).

A Session ID created by the device is notified by the response of this command, and the Session ID is required by both `Data Transfer Setup/Get` commands.

Table 3-81 Get Panic Log command

Command ID	ADK release	Variants
0x03	20.2	All

Command parameters

None

Table 3-82 Get Panic Log Response PDU contents

Payload byte	Value	Comments
1	(any)	Session ID (MSB)
2	(any)	Session ID (LSB)
3	(any)	Size of panic log in bytes (MSB)
4	(any)	Size of panic log in bytes (Second SB)
5	(any)	Size of panic log in bytes (Third SB)
6	(any)	Size of panic log in bytes (LSB)

Table 3-83 Get Panic Log command possible error status codes

Status	Code	Description
No data	0x81	The debug partition is empty.
Invalid parameters	0x85	Invalid parameters(s).
No debug partition	0x86	The debug partition is not found. (The partition must be defined in the flash layout config file.)
Failure by unknown reason	0x8F	Unspecified reason caused the failure of the requested operation.

3.4.4 Erase Panic Log command

The `Erase Log Info` command erases the entire debug partition.

Table 3-84 Erase Panic Log command

Command ID	ADK release	Variants
0x04	20.2	All

Table 3-85 Erase Panic Log command parameters

Payload byte	Value	Comments
1	0x00	Reserved for Future Use (MSB)
2	0x00	Reserved for Future Use (LSB)

These parameters must be set to zero to erase the `debug_partition`.

Response PDU Contents

None

Table 3-86 Erase Panic Log command possible error status codes

Status	Code	Description
Busy	0x83	The resource (the debug partition) is busy.
Invalid parameters	0x85	Invalid Debug Log Info Key.
No debug partition	0x86	The debug partition is not found. (The partition must be defined in the flash layout config file.)
Failure by unknown reason	0x8F	Unspecified reason caused the failure of the requested operation.

3.4.5 Debug Tunnel To Chip command

The `Debug Tunnel To Chip` command provides a tunnel between the device and the mobile app.

Table 3-87 Debug Tunnel To Chip command

Command ID	ADK release	Variants
0x05	20.2	All

Table 3-88 Debug Tunnel To Chip command parameters

Payload byte	Value	Comments
1	(any)	Client ID
2	(any)	Tag
3	(any)	Tunnelling payload (1/N)
...	...	...
N + 2	(any)	Tunnelling payload (N/N)

Table 3-89 Debug Tunnel To Chip response PDU contents

Payload byte	Value	Comments
1	(any)	Client ID
2	(any)	Tag
3	(any)	Tunnelling payload (1/N)
...	...	...
N + 2	(any)	Tunnelling payload (N/N)

Table 3-90 Debug Tunnel To Chip command possible error status codes

Status	Code	Description
Invalid parameters	0x85	Invalid parameter(s).
Failure by unknown reason	0x8F	Unspecified reason caused the failure of the requested operation.

3.4.6 GAIA Debug Error status code

The GAIA Debug feature commands could return error responses with the status code below.

Table 3-91 GAIA Debug error status codes

Status	Code	Description
No data	0x81	The debug partition is empty.
Not enough space	0x82	No space is available to store panic log.
Busy	0x83	The resource (the debug partition) is busy.
Invalid command	0x84	Invalid or not supported command ID.
Invalid parameters	0x85	Invalid parameters.
No debug partition	0x86	The debug partition is not found.
Unknown error	0x8F	Unspecified reason caused the failure of the requested operation.

If the mobile app sends a `GAIA Debug feature` command that is invalid or not supported, instead of the GAIA Debug feature, the GAIA Framework sends the error response to the mobile app with the error code of `0x01 Command not supported`.

3.5 Music Processing feature

The Music Processing GAIA feature is enabled by default on Headset Applications but not Earbud Applications. To enable this feature on the Earbud Application, add into the project definitions:

```
#define INCLUDE_MUSIC_PROCESSING
#define INCLUDE_MUSIC_PROCESSING_PEER
```

3.5.1 Get EQ State command

The GAIA client uses the `Get EQ State` command to decide whether the user can interact with the User EQ settings (predefined or user set).

The response contains whether the User EQ is present in the running audio chain or not.

Table 3-92 Get EQ State command

Command ID	Version	ADK release	Variants
0	1	20.3	All

Command parameters

None

Table 3-93 Get EQ State parameters

Payload byte	Value	Comments
1	0 or 1	0: Not present in chain 1: Present in chain

Possible Feature specific Error Status codes

None

3.5.2 Get Available EQ Pre-sets command

GAIA client uses the `Get Available EQ Pre-sets` command to find out the IDs of the supported presets. Each preset is identified by a number, which you must convert to a string and present to the user.

Table 3-94 Get Available EQ Pre-sets command

Command ID	Version	ADK release	Variants
1	1+	20.3	All

The response contains a list of IDs supported by the application. This set will include the Off, User and the defined preset IDs. These IDs will be used by other commands/notifications

Command Parameters

None

Table 3-95 Get Available EQ response PDU contents

Payload byte	Value	Comments
1	0 to 63	Number of presets (including off and user)
2	0 to 63	0: Off
3	0 to 63	1: Pre-set
...	...	...
n	0 to 63	3F: User

Possible feature-specific Error Status codes

None

3.5.3 Get EQ Set command

GAIA client uses the `Get EQ Set` command to find out what the currently selected preset (or User or off) is.

Table 3-96 Get EQ Set command

Command ID	Version	ADK release	Variants
2	1+	20.3	All

The response contains Off, User Set or Pre-set ID. Off means the EQ is in pass-through.

Command Parameters

None

Table 3-97 Get EQ Set response PDU contents

Payload byte	Value	Comments
1	0 to 63	0: Off 1: Pre-set 3F: User

Possible feature-specific Error Status codes

None

3.5.4 Set EQ Set command

GAIA client uses the `Set EQ Set` command to set the new preset value or user set or Off.

Table 3-98 Set EQ Set command

Command ID	Version	ADK release	Variants
3	1+	20.3	All

This command contains Off, User Set or Pre-set ID. Off sets the EQ into pass-through.

Table 3-99 Set EQ Set command parameters

Payload byte	Value	Comments
1	0 to 63	▪ 0: Off ▪ 1: Pre-set ▪ 3F: User

Table 3-100 Set EQ Set response PDU contents

Payload byte	Value	Comments
1	1	Ack

Possible feature-specific Error Status codes

None

3.5.5 Get User Set Number Of Bands command

GAIA client uses the `Get User Set Number of Bands` command to find out how many frequency bands the user set supports.

Table 3-101 Get User Set Number of Bands command

Command ID	Version	ADK release	Variants
4	1+	20.3	All

Command Parameters

None

Table 3-102 Get User Set Number of Bands response PDU contents

Payload byte	Value	Comments
1	0 to 9	Number of bands

Possible feature-specific Error Status codes

None

3.5.6 Get User Set Configuration command

GAIA client uses the `Get User Set Configuration` command to find out the details of each band.

Table 3-103 Get User Set Configuration command

Command ID	Version	ADK release	Variants
5	1+	20.3	All

Table 3-104 Get User Set Configuration command parameters

Payload byte	Value	Comments
1	0 to 9	Start band
2	0 to 9	End band

Table 3-105 Get User Set Configuration Response PDU contents

Payload byte	Value	Comments
1	0 to 9	Start band
2	0 to 9	End band
3 to 4	0 to 65535	Centre frequency
5-6	0 to 65535	Q
7	0 to 255	Filter type
8 to 9	0 to 65535	Gain value
...	...	...
66 to 67	0 to 65535	Centre frequency
68 to 69	0 to 65535	Q
70	0 to 255	Filter type
71 to 72	0 to 65535	Gain value

Possible-feature specific Error Status codes

None

3.5.7 Set User Set Configuration command

GAIA client uses the `Set User Set Configuration` command to set the gains of a specific set of bands.

Table 3-106 Set User Set Configuration command

Command ID	Version	ADK release	Variants
6	1+	20.3	All

Table 3-107 Set User Set command parameters

Payload byte	Value	Comments
1	0 to 9	Start band
2	0 to 9	End band
3 to 4	0 to 65535	Gain value
...	...	...
23 to 24	0 to 65535	Gain value

Table 3-108 Set User Set Response PDU contents

Payload byte	Value	Comments
1	1	Ack

Possible feature-specific Error Status codes

None

3.5.8 EQ State Change notification

GAIA client will be told if the User EQ is not present (for example, if a headset was streaming music so the User EQ was present but the phone started ringing which removes the User EQ).

Table 3-109 EQ Change notification

Notification ID	Version	ADK release	Variants
0	1+	20.3	All

Table 3-110 EQ State Change notification parameters

Payload byte	Value	Comments
1	0 or 1	■ 0: Not present ■ 1: Present

3.5.9 EQ Set Change notification

GAIA client will be told if the User EQ set (preset, User set or Off) changes.

Table 3-111 EQ Set Change notification

Notification ID	Version	ADK Release	Variants
1	1+	20.3	All

Table 3-112 EQ Set Change notification parameters

Payload byte	Value	Comments
1	0 to 63	■ 0: Off ■ 1: Pre-set ■ 3F: User

3.5.10 User EQ Band Change notification

This notification tells the GAIA client if there are User EQ band changes.

Table 3-113 User EQ Band Change notification

Notification ID	Version	ADK release	Variants
2	1+	20.3	All

Table 3-114 User EQ Band Change notification parameters

Payload byte	Value	Comments
1	0-9	Number of bands changed
2	0-9	Changed band number
...	...	...
10	0-9	Changed band number

3.6 Device Firmware Upgrade (DFU) feature

Table 3-115 DFU commands

Feature ID	Version	ADK release
0x06	1	20.1
	2	20.3

3.6.1 Upgrade Connect command

The `Upgrade Connect` command connects a GAIA transport to the upgrade library.

Table 3-116 Upgrade Connect command

Command ID	Version	ADK release	Variants
0	1+	20.1	All

Command parameters

None

Response PDU contents

None

Possible Feature specific Error Status codes

None

3.6.2 Upgrade Disconnect command

The `Upgrade Disconnect` command disconnects a GAIA transport from the upgrade library.

Table 3-117 Upgrade Disconnect command

Command ID	Version	ADK release	Variants
1	1+	20.1	All

Command parameters

None

Response PDU contents

None

Possible Feature specific Error Status codes

None

3.6.3 Upgrade Control command

The `Upgrade Control` command tunnels Upgrade messages to the device.

Table 3-118 Upgrade Control command

Command ID	Version	ADK release	Variants
2	1+	20.1	All

Command parameters

None

Response PDU contents

None

Possible Feature specific Error Status codes

None

3.6.4 Get Data Endpoint Mode command

The `Get Data Endpoint Mode` command returns the data endpoint that is set.

Table 3-119 Get Data Endpoint Mode command

Command ID	Version	ADK release	Variants
3	1+	20.1	All

Command parameters

None

Table 3-120 Get Data Endpoint Mode Response PDU parameters

Payload byte	Value	Comments
1	0 or 1	Data endpoint mode: ■ 0 : None ■ 1: RWCP

Possible Feature specific Error Status codes

None

3.6.5 Set Data Endpoint Mode command

The `Set Data Endpoint Mode` command sets the endpoint to use for sending data during an upgrade.

Table 3-121 Set Data Endpoint Mode command

Command ID	Version	ADK release	Variants
4	1+	20.1	All

Table 3-122 Set Data Endpoint Mode command parameters

Payload byte	Value	Comments
1	0 or 1	Data endpoint mode: ■ 0: None ■ 1: RWCP

Response PDU contents

None

Possible Feature-specific Error Status codes

None

3.6.6 Upgrade Data Indication notification

The Upgrade Data Indication notification tunnels Upgrade messages from the device.

Table 3-123 Upgrade Data Indication notification

Notification ID	Version	ADK release	Variants
0	1+	20.1	All

Table 3-124 Upgrade Data Indication notification parameters

Payload byte	Value	Comments
1	Data	Upgrade data

3.6.7 Upgrade Stop Request notification

Upgrade Stop Request notification indicates that device would like upgrade to stop.

Table 3-125 Upgrade Stop Request notification

Notification ID	Version	ADK release	Variants
0	2+	20.3	All

Table 3-126 Upgrade Stop Request notification parameters

Payload byte	Value	Comments
1	Action	0x00: Disconnect upgrade 0x01: Stop sending data (pause)

3.6.8 Upgrade Start Request notification

This notification indicates that device is ready to restart or resume an upgrade.

Table 3-127 Upgrade Start Request notification

Notification ID	Version	ADK release	Variants
0	2+	20.3	All

Table 3-128 Upgrade Start Request notification parameters

Payload byte	Value	Comments
1	Action	0x00: (Re)connect upgrade 0x01: Start sending data (resume)

3.7 Handset service feature

3.7.1 Enable Multipoint command

The GAIA Client uses this command to switch between single-point and multi-point.

Table 3-129 Enable Multipoint command

Command ID	Version	ADK Release	Variants
0	1	TBD	All

Table 3-130 Enable Multipoint command parameters

Payload byte	Value	Comments
1	0 or 1	0 = Single-point 1 = Multi-point

Response PDU contents

None

Possible feature specific error status codes

None

3.7.2 Multipoint Enabled Changed notification

The GAIA Client will be told if multipoint has been enabled or disabled.

Table 3-131 Multipoint Enabled Changed notification

Notification ID	Feature version	ADK release	Variants
0	1+	TBD	All

Table 3-132 Multipoint Enabled Changed notification parameters

Payload byte	Value	Comments
1	0 or 1	0 = Disabled 1 = Enabled

3.8 Audio Curation (AC) feature

Table 3-133 AC commands

Feature ID	Version	ADK release
0x08	1	21.1

3.8.1 Get State command

The `Get State` command provides the current state of the device Audio Curation feature type.

Table 3-134 Get State command

Command ID	Feature version	ADK release	Variants
0	1	21.1	All

Table 3-135 Get State command parameters

Payload byte	Value	Comments
1		Feature type
	0x01	ANC

Response PDU contents

Returns the state of the AC feature type.

Table 3-136 Get State response parameters

Payload Byte	Value	Comments
1		Feature type
	0x01	ANC

Table 3-136 Get State response parameters (cont.)

Payload Byte	Value	Comments
2	0x00	Disable
	0x01	Enable

3.8.2 Set State command

The `Set State` command allows the AC feature type to be turned On and Off.

Table 3-137 Set State command

Command ID	Feature version	ADK release	Variants
1	1	21.1	All

Table 3-138 Set State command parameters

Payload byte	Value	Comments
1		Feature type
	0x01	ANC
2	0x00	Disable
	0x01	Enable

Response PDU contents

None

Possible Feature-specific Error status codes

None

3.8.3 Get Modes Count command

The device returns the number of modes configured in the device. Modes are a predefined set of filter coefficients and related parameters for audio curation.

Table 3-139 Get Modes Count command

Command ID	Feature version	ADK release	Variants
2	1	21.1	All

Command parameters

None

Response PDU contents

Returns the number modes configured in the device.

Table 3-140 Get Num Modes response parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Number of configured modes

Possible Feature-specific Error status codes

None

3.8.4 Get Current Mode command

The device returns the configured mode currently set.

Table 3-141 Get Current ANC Mode command

Command ID	Feature version	ADK release	Variants
3	1	21.1	All

Command parameters

None

Response PDU contents

Returns the current mode and associated mode information.

Table 3-142 Get Current ANC Mode response parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0xff	Feature type: 0x01: Static ANC 0x02: Leakthrough ANC 0x03: Adaptive ANC 0x04 to 0xff: Reserved
3	0x00 to 0x01	Adaptation control support: 0x00: Not supported 0x01: Supported
4	0x00 to 0x01	Gain control support: 0x00: Not supported 0x01: Supported

3.8.5 Set Mode command

The `Set Mode` command configures the device to the set filter coefficients that corresponds to the provided Mode ID.

Table 3-143 Set Mode command

Command ID	Feature version	ADK release	Variants
4	1	21.1	All

Table 3-144 Set Mode command parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Mode to set

Response PDU contents

None

3.8.6 Get Gain command

The `Get Gain` command returns the gain of the device.

Table 3-145 Get Gain command

Command ID	Feature version	ADK release	Variants
5	1	21.1	All

Command parameters

None

Table 3-146 Get Gain response PDU contents

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0xff	Feature type: 0x01: Static ANC 0x02: Leakthrough ANC 0x03: Adaptive ANC 0x04 to 0xff: Reserved
3	0x00 to 0xff	Left gain value
4	0x00 to 0xff	Right gain value

Table 3-147 Get Gain possible Error status code

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.7 Set Gain command

The `Set Gain` command sets the gain of the device to the provided value. This is only applicable to modes where gain change is allowed by the user, for example, Leakthrough ANC modes.

Table 3-148 Set Gain command

Command ID	Feature version	ADK release	Variants
6	1	21.1	All

Table 3-149 Set Gain command parameters

Payload byte	Value	Comments
1	0x00 to 0xff	Left gain value
2	0x00 to 0xff	Right gain value

- NOTE**
- (1) The gain values can be changed for Leakthrough mode. The gain values must be the same for Left and Right in this case. Returns invalid parameter if the values are different.
 - (2) The gain values cannot be changed for Static and Adaptive modes. The device returns Incorrect State if requested for these modes.

Response PDU contents

None

Table 3-150 Set Gain possible Error status code

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.8 Get Toggle Configuration Count command

The `Get Toggle Configuration Count` command provides the number of toggle configurations the device supports.

Table 3-151 Get Toggle Configuration Count command

Command ID	Feature version	ADK release	Variants
7	1	21.1	RDP

Command parameters

None

Table 3-152 Get Toggle Configuration Count response PDU contents

Payload byte	Value	Comments
1	0x00 to 0x7f	Number of toggle options that are supported.

3.8.9 Get Toggle Configuration command

The `Get Toggle Configuration` command gets the option that is set for a toggle configuration.

Table 3-153 Get Toggle Configuration command

Command ID	Feature version	ADK release	Variants
8	1	21.1	All

Table 3-154 Get Toggle Configuration command parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Toggle option number

Table 3-155 Get Toggle Configuration response PDU contents

Payload byte	Value	Comments
1	0x01 to 0x7f	Toggle option number
2	0x00 to 0xff	Toggle option value: 0x00: Off 0x01 to 0x7f: Configured mode 0x80 to 0xfe: Reserved 0xff: Void/Toggle option not configured

Table 3-156 Get Toggle Configuration possible Error status code

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.10 Set Toggle Configuration command

The `Set Toggle Configuration` command sets the toggle option the user can toggle using controls on the device.

Table 3-157 Set Toggle Configuration command

Command ID	Feature version	ADK release	Variants
9	1	21.1	All

Table 3-158 Set Toggle Configuration command parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	ANC Toggle option number
2	0x00 to 0xff	ANC Toggle option value: 0x00: Off 0x01 to 0x7f: Configured mode 0x80 to 0xfe: Reserved 0xff: Void/Not connected

Response PDU contents

None

Table 3-159 Set Toggle Configuration possible Error status code

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.11 Get Scenario Configuration command

The `Get Scenario Configuration` command gets the behavior configured for a specific scenario on the device.

Table 3-160 Get Scenario Configuration command

Command ID	Feature version	ADK release	Variants
10	1	21.1	All

Table 3-161 Get Scenario Configuration command parameters

Payload byte	Value	Comments
1	0x00 to 0xff	Scenario: 0x01: Idle 0x02: Playback/Music 0x03: Voice call 0x04: Digital assistant 0x05: Reserved

Table 3-162 Get Scenario Configuration response PDU contents

Payload byte	Value	Comments
1	0x00 to 0xff	Scenario: 0x01: Idle 0x02: Playback/Music 0x03: Voice call

Table 3-162 Get Scenario Configuration response PDU contents (cont.)

Payload byte	Value	Comments
		0x04: Digital assistant 0x05: Reserved
2	0x01 to 0xff	Behavior: 0x00: Off 0x01 to 0x7f: Configured mode 0x80 to 0xfe: Reserved 0xff: Same as current

Table 3-163 Get Scenario Configuration possible Error status code

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.12 Set Scenario Configuration command

The `Set Scenario Configuration` command sets the behavior to be used when in different scenarios, for example, idle state, voice calls or digital assistant use.

Table 3-164 Set Scenario Configuration command

Command ID	Feature version	ADK release	Variants
11	1	21.1	All

Table 3-165 Set Scenario Configuration command parameters

Payload byte	Value	Comments
1	0x00 to 0xff	Scenario: 0x01: Idle 0x02: Playback/Music 0x03: Voice call 0x04: Digital assistant 0x05: Reserved
2	0x01 to 0xff	Behavior: 0x00: Off 0x01 to 0x7f: Configured mode 0x80 to 0xfe: Reserved 0xff: Same as current

Response PDU contents

None

Table 3-166 Set Scenario Configuration possible Error status code

Status	Code	Description
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.13 Get Demo Support command

The `Get Demo Support` command is used to identify if the device supports demonstration state.

Table 3-167 Get Demo Support command

Command ID	Feature version	ADK release	Variants
12	1	21.1	RDP

Command parameters

None

Table 3-168 Get Demo Support response PDU contents

Payload byte	Value	Comments
1	0x00	Not supported
	0x01	Supported

3.8.14 Get Demo State command

The `Get Demo State` command is used to identify if the device is currently in a demonstration state.

Table 3-169 Get Demo State command

Command ID	Feature version	ADK release	Variants
13	1	21.1	RDP

Command parameter

None

Table 3-170 Get Demo State command response PDU contents

Payload byte	Value	Comments
1	0x00	Out of Demo mode
	0x01	In Demo mode

3.8.15 Set Demo State command

The `Set Demo State` command lets the device know if AC is to be moved into or out of demo. In demo user has more options to control the AC behavior allowing them to get a flavor of AC functionality supported.

Device can move out of demo mode on GAIA disconnect.

Table 3-171 Set Demo State command

Command ID	Feature version	ADK release	Variants
14	1	21.1	RDP

Table 3-172 Set Demo Support command parameters

Payload byte	Value	Comments
1	0x00	Exit Demo mode
	0x01	Enter Demo mode

Response PDU contents

None

Table 3-173 Set Demo State command possible error status codes

Status	Code	Description
Incorrect state	0x06	The device is not the correct state to process the command.

3.8.16 Get Adaptation Control Status command

The Get Adaptation Control Status command gets the state of gain adaptation.

It is only applicable if the device is in Demo mode.

Table 3-174 Get Adaptation Control Status command

Command ID	Feature version	ADK release	Variants
15	1	21.1	All

Command parameters

None

Table 3-175 Get Adaptation Control Status response PDU contents

Payload byte	Value	Comments
1	0x00	Adaptation suspended
	0x01	Adaptation running

Table 3-176 Get Adaptation Control Status possible error status codes

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.

3.8.17 Set Adaptation Control Status command

The `Set Adaptation Control Status` command sets the state of gain adaptation.

It is only applicable if the device is in Demo mode.

Table 3-177 Set Adaptation Control Status command

Command ID	Feature version	ADK release	Variants
16	1	21.1	All

Table 3-178 Set Adaptation Control Status command parameters

Payload byte	Value	Comments
1	0x00	Pause/Suspend adaptation
	0x01	Resume adaptation

Response PDU contents

None

Table 3-179 Set Adaptation Control Status possible error status codes

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.

3.8.18 Get Leakthrough dB Gain Slider Configuration command

This command provides the slider configuration supported by device for current mode. It gets the number of steps in which leakthrough dB gain can be configured, dB value of each step, minimum dB gain that can be configured and the step corresponding to current leakthrough dB gain.

Table 3-180 Get Leakthrough dB Gain Slider Configuration command

Command ID	Feature version	ADK release	Variants
17	2	21.4	All

Command parameters

None

Table 3-181 Get Leakthrough dB Gain Slider Configuration command response PDU contents

Payload byte	Value	Comments
1	0x01 to 0x7F	Current mode
2	0x03 to 0x0A	Number of steps that are supported

Table 3-181 Get Leakthrough dB Gain Slider Configuration command response PDU contents (cont.)

Payload byte	Value	Comments																					
3	0x02 to 0x05	dB step size in which leakthrough gain can be updated																					
4	0x00 to 0xFF	Minimum leakthrough gain supported by device. Payload represents an 8-bit signed representation of minimum leakthrough dB gain (-128 dB to 127 dB) <table> <tr> <th>Payload (bin)</th><th>Payload (hex)</th><th>dB gain</th></tr> <tr> <td>0000 0000</td><td>0x00</td><td>0</td></tr> <tr> <td>0000 1010</td><td>0x0A</td><td>10</td></tr> <tr> <td>0111 1111</td><td>0x7F</td><td>127</td></tr> <tr> <td>1000 0000</td><td>0x80</td><td>-128</td></tr> <tr> <td>1111 0110</td><td>0xF6</td><td>-10</td></tr> <tr> <td>1111 1111</td><td>0xFF</td><td>-1</td></tr> </table>	Payload (bin)	Payload (hex)	dB gain	0000 0000	0x00	0	0000 1010	0x0A	10	0111 1111	0x7F	127	1000 0000	0x80	-128	1111 0110	0xF6	-10	1111 1111	0xFF	-1
Payload (bin)	Payload (hex)	dB gain																					
0000 0000	0x00	0																					
0000 1010	0x0A	10																					
0111 1111	0x7F	127																					
1000 0000	0x80	-128																					
1111 0110	0xF6	-10																					
1111 1111	0xFF	-1																					
5	0x01 to 0x0A	Current step at which leakthrough slider points to																					

- NOTE**
1. This command is only applicable for static/adaptive leakthrough mode. The device returns incorrect State if requested for other mode configurations.
 2. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices.
 3. This configuration may differ for each leakthrough mode
 4. Leakthrough dB gain can be obtained from current step using formula: $\text{MinGain} = ((\text{CurStep} - 1) * \text{StepSize})$

Table 3-182 Get Leakthrough dB Gain Slider Configuration command possible feature specific error status codes

Status	Code	Description
Incorrect state	0x06	This device is not in the correct state to process the command.

3.8.19 Get Current Leakthrough dB Gain Step command

The `Get Current Leakthrough dB Gain Step` command provides the current step corresponding to leakthrough gain with which device is configured.

Table 3-183 Get Current Leakthrough dB Gain Step command

Command ID	Feature version	ADK release	Variants
18	2	21.4	All

Command parameters

None

Table 3-184 Get Current Leakthrough dB Gain Step command response PDU contents

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0x0A	Current step at which leakthrough slider point to

- NOTE**
1. This command is only applicable for static/adaptive leakthrough mode. The device returns incorrect State if requested for other mode configurations.
 2. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices
 3. Leakthrough dB gain can be obtained from current step using formula: $\text{MinGain} + ((\text{CurStep} - 1) * \text{StepSize})$

Table 3-185 Get Current Leakthrough dB Gain Step command response PDU contents

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.

3.8.20 Set Leakthrough dB Gain Step command

The Set Leakthrough dB Gain Step command sets the leakthrough gain on the device corresponding to step value provided by user.

Table 3-186 Set Leakthrough dB Gain Step command

Command ID	Feature version	ADK release	Variants
19	2	21.4	All

Table 3-187 Set Leakthrough dB Gain Step command parameters

Payload byte	Value	Comments
1	0x01 to 0x0A	Step value to which leakthrough gain is to be updated

Response PDU contents

None

- NOTE**
1. If the step value exceeds maximum supported value or goes below minimum supported value, device returns Invalid Parameter error code.
 2. This command is only applicable for static/adaptive leakthrough mode. The device returns Incorrect State if requested for other mode configurations.
 3. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices.
 4. Leakthrough dB gain can be obtained from step using formula: $\text{MinGain} + (\text{Step} - 1) * \text{StepSize}$

Table 3-188 Set Leakthrough dB Gain Step command possible feature specific error status codes

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.
Invalid parameter	0x05	An invalid parameter was used in the command.

3.8.21 Get Left Right Balance command

The `Get Left Right Balance` command provides the current setting of balance with which device is configured.

Table 3-189 Get Left Right Balance command

Command ID	Feature version	ADK release	Variants
20	2	21.4	All

Command parameters

None

Table 3-190 Get Left Right Balance command response PDU contents

Payload byte	Value	Comments
1	0x00	Left
	0x01	Right
2	0x00 to 0x064	Gain

- NOTE**
1. This command is only applicable for static/adaptive leakthrough mode. The device returns Incorrect State if requested for other mode configurations.
 2. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices.
 3. This balance remains constant across all leakthrough modes.

Table 3-191 Get Left Right Balance command possible feature specific error status codes

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.

3.8.22 Set Left Right Balance command

The Set Left Right Balance command sets the balance of the device to the provided value. This is only applicable to modes where gain change is allowed by the user, for example, Static/Adaptive Leakthrough ANC modes. This balance will remain constant across all leakthrough modes.

Table 3-192 Set Left Right Balance command

Command ID	Feature version	ADK release	Variants
21	2	21.4	All

Table 3-193 Set Left Right Balance command parameters

Payload byte	Value	Comments
1	0x00	Left
	0x01	Right
2	0x00 to 0x064	Gain

Response PDU contents

None

- NOTE**
1. The gain values must be in the range of 0 to 100 defining the percentage of balance. Returns invalid parameter if the values are different.
 2. The gain values cannot be changed for Static and Adaptive modes. The device returns Incorrect State if requested for these modes.

Table 3-194 Set Left Right Balance command possible feature specific error status codes

Status	Code	Description
Incorrect state	0x06	The device is not in the correct state to process the command.
Invalid parameter	0x05	An invalid parameter was used in the command

3.8.23 Get Wind Noise Reduction Support command

This command is used to identify if the device supports Wind Noise Reduction feature.

Table 3-195 Get Wind Noise Reduction Support command

Command ID	Feature version	ADK release	Variants
22	2	21.4	QCC517x/307x

Command parameters

None

Table 3-196 Get Wind Noise Reduction Support command response PDU contents

Payload byte	Value	Comments
1	0x00	Not supported
	0x01	Supported

Possible feature specific error codes

None

3.8.24 Get Wind Noise Detection State command

This command provides the current state of the Wind Noise Reduction feature on device.

Table 3-197 Get Wind Noise Detection State command

Command ID	Feature version	ADK release	Variants
23	2	21.4	QCC517x/307x

Command parameters

None

Response PDU contents

Returns the state of the Wind Noise Reduction feature state.

Table 3-198 Get Wind Noise Detection State command response PDU contents

Payload byte	Value	Comments
1	0x00	Disable
	0x01	Enable

Possible feature specific error status codes

None

3.8.25 Set Wind Noise Detection State command

This command allows the device to turn Wind Noise Reduction feature state On or Off.

Table 3-199 Set Wind Noise Detection State command

Command ID	Feature version	ADK release	Variants
24	2	21.4	QCC517x/307x

Table 3-200 Set Wind Noise Detection State command parameters

Payload byte	Value	Comments
1	0x00	Disable wind noise detection
	0x01	Enable wind noise detection

Response PDU contents

None

Possible feature specific error status codes

None

3.8.26 State Change notification

The State Change notification is sent when the AC feature type state changes.

Table 3-201 State Change notification

Notification	Feature version	ADK release	Variants
0	1	21.1	All

Table 3-202 State Change notification parameters

Payload byte	Value	Comments
1		Feature type
	0x01	ANC
2	0x00	Disabled
	0x01	Enabled

3.8.27 Mode Change notification

The Mode Change notification is sent when the AC mode is updated.

Table 3-203 Mode Change notification

Notification	Feature version	ADK release	Variants
1	1	21.1	All

Table 3-204 Mode Change notification parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0xff	Feature type: ■ 0x01: Static ANC ■ 0x02: Leakthrough ANC ■ 0x03: Adaptive ANC ■ 0x04 to 0xff: Reserved

Table 3-204 Mode Change notification parameters (cont.)

Payload byte	Value	Comments
3	0x00 to 0x01	Adaptation control support: 0x00: Not supported 0x01: Supported
4	0x00 to 0x01	Gain control support: 0x00: Not supported 0x01: Supported

3.8.28 Gain Change notification

The Gain Change notification indicates the currently configured AC modes the user can toggle using the button/touch sensor of the device.

Table 3-205 Gain Change notification

Notification	Feature version	ADK release	Variants
2	1	21.1	All

Table 3-206 Gain Change notification parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0xff	Feature type: 0x01: Static ANC 0x02: Leakthrough ANC 0x03: Adaptive ANC 0x04 to 0xff: Reserved
3	0x00 to 0xff	Left gain value
4	0x00 to 0xff	Right gain value

3.8.29 Toggle Configuration notification

The Toggle Configuration notification indicates the currently configured AC modes the user can toggle using the button/touch sensor of the device.

Table 3-207 Toggle Configuration notification

Notification	Feature version	ADK release	Variants
3	1	21.1	All

Table 3-208 Toggle Configuration notification parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Toggle Configuration number
2	0x00	Off
	0x01 to 0x7f	AC mode
	0xff	Void/ Toggle option not configured

3.8.30 Scenario Configuration notification

The Scenario Configuration notification indicates the AC mode behavior when in different scenarios, like idle state, voice calls or digital assistant use.

Table 3-209 Scenario Configuration notification

Notification	Feature version	ADK release	Variants
4	1	21.1	All

Table 3-210 Scenario Configuration notification parameters

Payload byte	Value	Comments
1	0x01 to 0x7f	Scenario: 0x01: Idle 0x02: Playback/Music 0x03: Voice call 0x04: Digital assistant 0x05: Reserved
2	0x00	Toggle behavior: 0x00: Off 0x01 to 0x7f: Configured mode 0x80 to 0xfe: Reserved 0xff: Same as current

3.8.31 Demo State notification

The Demo State notification indicates if the device is in or out of Demo mode.

Table 3-211 Demo State notification

Notification	Feature version	ADK release	Variants
5	1	21.1	RDP

Table 3-212 Demo State notification parameters

Payload byte	Value	Comments
1	0x00	Out of Demo mode
	0x01	In Demo mode

3.8.32 Adaptation Status Change notification

The Adaptation Status Change notification is sent if the Adaptation Status is paused or resumed.

Table 3-213 Adaptation Status Change notification

Notification	Feature version	ADK release	Variants
6	1	21.1	All

NOTE (1) Applicable in Demo mode.

(2) Applicable to Adaptive ANC modes only.

Table 3-214 Adaptation Status Change notification parameters

Payload byte	Value	Comments
1	0x00	Paused
	0x01	Resumed

3.8.33 Leakthrough dB Gain Slider Configuration notification

This notification indicates the slider configuration supported by device for current mode. The notification provides the number of steps in which leakthrough dB gain can be configured, dB value of each step, minimum dB gain that can be configured and the step corresponding to current leakthrough dB gain.

Table 3-215 Leakthrough dB Gain Slider Configuration notification

Notification	Feature version	ADK release	Variants
7	2	21.4	All

Table 3-216 Leakthrough dB Gain Slider Configuration notification parameters

Payload byte	Value	Comments
1	0x01 to 0x7F	Current mode
2	0x03 to 0x0A	Number of steps that are supported
3	0x02 to 0x05	dB step size in which leakthrough gain can be updated
4	0x00 to 0xFF	Minimum leakthrough dB gain supported by device

Table 3-216 Leakthrough dB Gain Slider Configuration notification parameters (cont.)

Payload byte	Value	Comments		
		Payload represents an 8-bit signed representation of minimum leakthrough dB gain (-128 dB to 127 dB)		
		Payload (bin)	Payload (hex)	dB gain
		0000 0000	0x00	0
		0000 1010	0x0A	10
		0111 1111	0x7F	127
		1000 0000	0x80	-128
		1111 0110	0xF6	-10
		1111 1111	0xFF	-1
5	0x01 to 0x0A	Current step at which leakthrough slider points to		

- NOTE**
1. This notification is only applicable for static/adaptive leakthrough mode.
 2. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices.
 3. This configuration may differ for each leakthrough mode.
 4. Leakthrough dB gain can be obtained from current step using formula: $\text{MinGain} + ((\text{CurStep} - 1) * \text{StepSize})$

3.8.34 Leakthrough dB Gain Change notification

The Leakthrough dB Gain Change notification is sent when leakthrough gain is updated and indicates the current step corresponding to leakthrough dB gain with which device is configured.

Table 3-217 Leakthrough dB Gain Change notification

Notification	Feature version	ADK release	Variants
8	2	21.4	All

Table 3-218 Leakthrough dB Gain Change notification

Payload byte	Value	Comments
1	0x01 to 0x7f	Current mode
2	0x01 to 0xA	Current step at which leakthrough slider should point to

- NOTE**
1. This notification is only applicable for static/adaptive leakthrough mode.
 2. Adaptive leakthrough mode is not supported on QCC514x, QCC515x, QCC304x, QCC305x devices.
 3. Leakthrough dB gain can be obtained from current step using formula: $\text{MinGain} + ((\text{CurStep} - 1) * \text{StepSize})$

3.8.35 Left Right Balance notification

This notification is sent when Left Right Balance is updated and indicates the current balance with which device is configured. This is only applicable to modes where gain change is allowed by the user, for example, Static/Adaptive Leakthrough ANC modes. This balance will remain constant across all leakthrough modes.

Table 3-219 Left Right Balance notification

Notification	Feature version	ADK release	Variants
9	2	21.4	All

Table 3-220 Left Right Balance notification parameters

Payload byte	Value	Comments
1	0x00	Left
	0x01	Right
2	0x00 to 0x064	Gain

NOTE The gain values will be in the range of 0 to 100 defining the percentage of balance.

3.8.36 Wind Noise Detection State change notification

This notification indicates the Wind Noise Reduction feature state of both earbuds.

Table 3-221 Wind Noise Detection State change notification

Notification	Feature version	ADK release	Variants
10	2	21.4	QCC517x/307x

Table 3-222 Wind Noise Detection State change notification parameters

Payload byte	Value	Comments
1	0x00	Disabled
	0x01	Enabled

3.8.37 Wind Noise Reduction Indication notification

This notification indicates the detection of wind noise on the left and right earbuds.

Table 3-223 Wind Noise Reduction Indication notification

Notification	Feature version	ADK release	Variants
11	2	21.4	QCC517x/307x

Table 3-224 Wind Noise Reduction Indication notification parameters

Payload byte	Value	Comments
1	0x00	Left: Wind not detected
	0x01	Left: Wind detected
2	0x00	Right: Wind not detected
	0x01	Right: Wind detected

3.9 Fit Status feature

Table 3-225 Fit Status

Feature ID	Version	ADK release
0x09	1	21.1

3.9.1 Set Fit Status command

The `Set Fit Status` command allows the device to perform the fit/wear check test.

Table 3-226 Set Fit Status command

Command ID	Feature version	ADK release	Variants
0	1	21.1	All

Table 3-227 Set Mode command parameters

Payload byte	Value	Comments
1	0x00	Stop Fit Status test
	0x01	Start Fit Status test

Response PDU contents

None

Possible Feature-specific Error status codes

None

3.9.2 Fit Status Indication notification

The Fit Status Indication notification indicates the fit status of the left and right earbud.

Table 3-228 Fit Status Indication notification

Notification	Feature version	ADK release	Variants
0	1	21.1	All

Table 3-229 Fit Status Indication notification parameters

Payload byte	Value	Comments
1	0x01 to 0x03	Left fit result: 0x01: Good fit 0x02: Bad fit 0x03: Fail
2	0x01 to 0x03	Right fit result: 0x01: Good fit 0x02: Bad fit 0x03: Fail
NOTE Good fit: The earbud can be considered as fitting well. Bad fit: The earbud is not fitting well. The earbud may need adjusting or a change of ear tip may be required. Fail: The fit test has failed. It could be due to various reasons, for example, user explicitly stopping the test or the test being interrupted due to a voice call or retesting needed in a quiet environment and so forth.		

3.10 Voice Processing feature

The Voice Processing feature is intended to demo the different Qualcomm® CVC™ features using the Qualcomm CVC 3Mic capability. The Voice Processing GAIA feature is disabled by default. To enable this feature, add into the project definition: `#define INCLUDE_CVC_DEMO`

3.10.1 Get Supported Enhancements command

The Gaia Client uses the `Get Supported Enhancements` command to decide whether the user can interact with the Voice Processing.

The response contains a list of capabilities that are available on the device.

Table 3-230 Get Supported Enhancements command

Command ID	Feature version	ADK release	Variants
0	1	21.3	All

Table 3-231 Get Supported Enhancements command parameters

Payload byte	Value	Comments
1	(any)	List offset: Transfer capabilities starting with this offset

Table 3-232 Get Supported Enhancements command response PDU contents

Payload byte	Value	Comments
1	0 or 1	0: No more data 1: More data available
2	0 or 1	1st enhancement: 0: CAPABILITY_NONE 1: CAPABILITY_CVC_3MIC

Possible feature specific error status codes

None

3.10.2 Set Config Enhancement command

The `Set Config Enhancement` command sets a mode for a specific capability.

Table 3-233 Set Config Enhancement command

Command ID	Feature version	ADK release	Variants
1	1	21.3	All

Table 3-234 Set Config Enhancement command parameters

Payload byte	Value	Comments
1	1	Select capability 1: CAPABILITY_CVC_3MIC
2	0 ~ 3	Microphone mode: 0: Bypass mode 1: cVc 1Mic mode 2: cVc 2Mic mode 3: cVc 3Mic mode
3	0 ~ 2	Bypass mode: 0: Bypass voice microphone 1: Bypass external microphone 2: Bypass internal microphone

Response PDU contents

None

Table 3-235 Set Config Enhancement command possible feature error status codes

Status	Code	Description
Invalid parameter	0x05	The parameter was invalid

3.10.3 Get Config Enhancement command

The `Get Config Enhancement` command reads out a mode for a specific capability.

Table 3-236 Get Config Enhancement command

Command ID	Feature version	ADK release	Variants
2	1	21.3	All

Table 3-237 Get Config Enhancement command parameters

Payload byte	Value	Comments
1	1	Select capability (1 = CAPABILITY_CVC_3MIC)

Table 3-238 GetConfigEnhancement command response PDU contents for capability CVC_3MIC

Payload byte	Value	Comments
1	1	Selected capability 1: CAPABILITY_CVC_2MIC
2	0 ~ 3	Microphone mode: 0: Bypass mode 1: cVc 1Mic mode 2: cVc 2Mic mode 3: cVc 3Mic mode
3	0 ~ 2	Bypass mode: 0: Bypass voice microphone 1: Bypass external microphone 2: Bypass internal microphone
4	0 ~ 1	Operation mode: 0: 2Mic mode 1: 3Mic mode 3Mic cVc capability operates in either 2-mic or 3-mic mode depending on the wind or noisy environment.

Table 3-239 Get Config Enhancement command possible feature specific error status codes

Status	Code	Description
Invalid parameter	0x05	The command was valid, but the device has no valid settings.
Incorrect state	0x06	The device is not in the correct state to process the command.

3.10.4 Notify Enhancement Mode Change notification

The Notify Enhancement Mode Change notification informs the GAIA app about changes in a specific capability.

The capability is defined in the first payload byte. The following values are specific to this capability.

Table 3-240 Notify Enhancement Mode Change notification

Notification ID	Feature version	ADK release	Variants
0	1	21.3	All

Table 3-241 Notify Enhancement Mode Change notification parameters

Payload byte	Value	Comments
1	1	Selected capability 1: CAPABILITY_CVC_3MIC
2	0 ~ 3	Microphone mode: 0: Bypass mode 1: cVc 1Mic mode 2: cVc 2Mic mode 3: cVc 3Mic mode
3	0 ~ 2	Bypass mode 0: Bypass voice microphone 1: Bypass external microphone 2: Bypass internal microphone
4	0 ~ 1	Operation mode: 0: 2Mic mode 1: 3Mic mode 3Mic cVc capability operates in either 2-mic or 3-mic mode depending on the wind or noisy environment.

Possible feature specific error status codes

None

3.11 UI Gesture Configuration feature

This feature allows the end user to configure what their Earbuds do when they perform gestures such as tap, swipe, long press and so on on the device touchpad sensor. It also allows differentiation of the gestures on the Left and Right Earbuds. For example, the end user may wish 'volume up' to be performed by a tap of the left Earbud and 'volume down' by a tap of the right Earbud.

The end user reconfigures the ROM default behavior using the 'Gesture configuration' screens of the iOS or Android GAIA Application. Their chosen settings are applied immediately when entered through the GAIA App GUI and stored in NVRAM on the devices. They persist through a DFU session. The list of configurable gestures, contexts and actions are designed to be extensible in future ADKs. Any

version of the GAIA App including this feature shall be compatible with any CAA Application that includes the feature.

NOTE This feature can be used with any CAA Application, not just Earbuds.

These commands and notifications are provided by the UI Configuration GAIA plug-in.

Table 3-242 UI Gesture Configuration

Feature ID	Version	ADK release
0x0b	1	21.3

3.11.1 Get Number of Touchpads command

The `Get Number of Touchpads` command is sent by the mobile application in order to discover the number of originating touchpads that the embedded device has available for configuration. This query has no message payload.

Table 3-243 Get Number of Touchpads command

Command ID	Version	ADK release	Variants
0	1+	21.4	All

Command parameters

None

Table 3-244 Get Number of Touchpads command response PDU contents

Payload byte	Value	Comments	
1	1 or 2	Value	Comment
		0	Reserved
		1	Denotes a device with a single originating touchpad. All Configuration Query/Set commands with this device have the originating touchpad value 0 == 'single'.
		2	Denotes a device in an Earbud pair or possibly a Headset application, where there are discrete Left and Right touchpads. Configuration Query/Set commands with this device have originating touchpad values of either 1 == 'right', 2 == 'left', or 3 == 'both'

Possible feature specific error status codes

None

3.11.2 Get Supported Gestures Command

The `Get Supported Gestures` command is sent by the mobile application in order to discover the Gesture IDs supported by the embedded device. This query has no message payload.

Table 3-245 Get Supported Gestures command

Command ID	Version	ADK release	Variants
1	1+	21.3	All

Command parameters

None

Table 3-246 Get Supported Gestures command response PDU contents

Payload byte	Value	Comments
1 to 16		Bit Array of supported Gesture IDs Gesture ID 0 is 1<<0, that is, 0x01 Gesture ID 7 is 1<<7, that is, 0x80 ... Gesture ID 127 is 0x80 in the 16th byte

This is a multi-byte bitfield, where the bit index denotes the Gesture ID. All supported gestures are set to 1. All unsupported gestures are set to 0.

Possible Feature specific Error Status codes

None

3.11.3 Get Supported Contexts command

The `Get Supported Contexts` command is sent by the mobile application in order to discover the Context IDs supported by the embedded device. This query has no message payload.

Table 3-247 Get Supported Contexts command

Command ID	Version	ADK release	Variants
2	1+	21.4	All

Command parameters

None

Table 3-248 Get Supported Contexts command response PDU contents

Payload byte	Value	Comments
1 to 16		Bit Array of supported Context IDs Context ID 0 is 1<<0, that is, 0x01 Context ID 7 is 1<<7, that is, 0x80 ... Context ID 127 is 0x80 in the 16th byte

This is a multi-byte bitfield, where the bit index denotes the Context ID. All supported contexts are set to 1. All unsupported contexts are set to 0.

Possible Feature specific Error Status codes

None

3.11.4 Get Supported Actions command

The `Get Supported Actions` command is sent by the mobile application in order to discover the Action IDs supported by the embedded device. This query has no message payload.

Table 3-249 Get Supported Actions command

Command ID	Version	ADK release	Variants
3	1+	21.3	All

Command parameters

None

Table 3-250 Get Supported Actions command response PDU contents

Payload byte	Value	Comments
1 to 16		Bit Array of supported Action IDs Action ID 0 is 1<<0, that is, 0x01 Action ID 7 is 1<<7, that is, 0x80 ... Action ID 127 is 0x80 in the 16th byte

This is a multi-byte bitfield, where the bit index denotes the Action Id. All supported actions are set to 1. All unsupported actions are set to 0.

Possible Feature specific Error Status codes

None

3.11.5 Get Configuration For Gesture command

This query is sent from the Mobile application to the UI Configuration GAIA plug-in. It is sent when the Mobile app needs to populate its GUI with the current UI configuration active in the embedded device.

The query comprises of two bytes:

1. Firstly, a byte specifies the Gesture ID whose configuration is being queried by the Mobile application.
2. Secondly, a byte specifies the offset for which to read the configuration data at. This is needed when the configuration data is indicated as exceeding the minimum 16-byte MTU (for some BLE transports) (this is signaled by the embedded app in its response to the first query for that gesture ID).

Table 3-251 Get Configuration For Gesture command

Command ID	Version	ADK release	Variants
4	1+	21.4	All

Table 3-252 Get Configuration For Gesture command parameters

Payload byte	Value	Comments
1		The Gesture ID for which the Mobile application is querying the configuration.
2		The byte offset to read the configuration data from.

Table 3-253 Get Configuration For Gesture command parameters

Payload byte	Value	Comments
1	Valid gesture ID	The Gesture ID for which the Mobile application is querying the configuration
2	Any	The byte offset to read the configuration from

Table 3-254 Get Configuration For Gesture command response PDU contents

Payload byte	Value	Comments																																
1	Any	Firstly, there is a single-byte bitfield specifying: <table><tr><th>Bit7</th><th>Bit6</th><th>Bit5</th><th>Bit4</th><th>Bit3</th><th>Bit2</th><th>Bit1</th><th>Bit0</th></tr><tr><td colspan="2">More data</td><td colspan="6">Gesture ID the configuration is for</td></tr></table> The more data flag is set to 1 if the embedded app has more data to send to the mobile app, that is, the payload needs to be fragmented over several packets. If this bit is 0, all the requested data is present in this response.	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	More data		Gesture ID the configuration is for																					
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0																											
More data		Gesture ID the configuration is for																																
2 and 3, 4 and 5, and so on (up to maximum of 14 and 15)	Any	Secondly, there is an array of an even number of bytes, where each byte pair forms a single 16 bit word. Therefore this is an array of 0 to 7 uint16s. Each uint16 represents a bitfield encoded as follows: <table><tr><th>Bit 5</th><th>Bit 4</th><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th><th>Bit 9</th><th>Bit 8</th><th>Bit 7</th><th>Bit 6</th><th>Bit 5</th><th>Bit 4</th><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr><tr><td colspan="2">Originating touchpad</td><td colspan="8">Configured Context ID</td><td colspan="6">Configured Action ID</td></tr></table>	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Originating touchpad		Configured Context ID								Configured Action ID					
Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0																			
Originating touchpad		Configured Context ID								Configured Action ID																								

Table 3-254 Get Configuration For Gesture command response PDU contents (cont.)

Payload byte	Value	Comments								
		The Origin Touchpad value denotes one of Single/Left/Right/Both and identifies the touchpad for which the gesture has this configured action and context. Single is used when there is only a single touchpad on the device. <table><tr><td>Single</td><td>0</td></tr><tr><td>Right</td><td>1</td></tr><tr><td>Left</td><td>2</td></tr><tr><td>Both</td><td>3</td></tr></table> Bits 7 to 13 denote the context ID that should match in order to trigger the configured action. Bits 0 to 6 denote the configured action ID.	Single	0	Right	1	Left	2	Both	3
Single	0									
Right	1									
Left	2									
Both	3									

Possible Feature specific Error Status codes

None

3.11.6 Set Configuration For Gesture command

This command is sent whenever the user changes the actions associated with a gesture in the Mobile App GUI. It uses a similar format to the response for the [Get Configuration For Gesture command](#) (that is, the message used to populate the Mobile Application's GUI with the original configuration state data). The mobile application shall send one or more commands (it may be necessary to modify the configuration of two or more gestures due to the users change) as follows:

Table 3-255 Set Configuration For Gesture command

Command ID	Version	ADK Release	Variants
5	1+	21.4	All

Table 3-256 Set Configuration For Gesture command parameters

Payload byte	Value	Comments								
1	Any	Firstly, there shall be a single-byte bitfield specifying:								
		<table><tr><td>Bit7</td><td>Bit6</td><td>Bit5</td><td>Bit4</td><td>Bit3</td><td>Bit2</td><td>Bit1</td><td>Bit0</td></tr></table>	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	
		<table><tr><td>More Data</td><td colspan="7">Gesture ID for which the configuration is being set</td></tr></table>	More Data	Gesture ID for which the configuration is being set						
More Data	Gesture ID for which the configuration is being set									
The <code>More Data</code> flag is set to 1 if the mobile app has more data to send to the plug-in, that is, the payload needs to be fragmented over several packets. If this bit is 0, all the data is presented in this command.										
2		Secondly, there is a byte specifying the index to write at (this is always 0 for the first packet of the Set command, and is a non-zero only when continuing fragmented SET commands which were spread over several packets).								

NOTE As 7 records (14 bytes) fit into a packet, the first continuation packet writes at index 7.

When the mobile App sends a new command (that is, to configure the next Gesture ID), then the index\once again starts at 0.

3 and 4,
5 and 6, and so on

Any

Thirdly, there is an array of an even number of bytes, where each byte pair forms a single 16-bit word. Therefore this is an array of 0 to 7 uint16s. Each uint16 represents a bitfield encoded as follows:

Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Originating Touchpad		Configured Context ID								Configured Action ID					

The originating touchpad value denotes one of Single/Left/Right/Both and identifies the touchpad which can originate the gesture that has this configured action and context.

Single	0
Right	1
Left	2
Both	3

Bits 7 to 13 denote the context ID that should match in order to trigger the configured action.

Bits 0 to 6 denote the configured action ID.

Response PDU contents

None

Possible Feature specific Error Status codes

None

3.11.7 Reset Configuration To Defaults command

This command is sent when the end user wishes to reset the embedded device(s) to their original default UI configuration from the GAIA application (that is, not via a factory reset of the entire device).

Table 3-257 Reset Configuration To Defaults command

Command ID	Version	ADK release	Variants
6	1+	21.4	All

Command Parameters

None

Response PDU contents

None

Table 3-258 Reset Configuration To Defaults command possible feature specific error status codes

Status	Code	Description
Peer Link Down	0x80	The removal of the NVRAM-stored end-user configuration can only proceed when the peer link between the Earbud devices is up at the time of the 'Reset Configuration to Defaults' command being received. If it is not up, the Primary Earbud will respond with this error code to the GAIA App. In that case the GAIA App presents a dialog to the user requesting them to ensure both Earbuds are out of case and ready for the configuration to be reset.

3.11.8 Gesture configuration Changed notification

The embedded application sends this notification to all connected mobile applications when a gesture configuration has been updated. This allows other connected applications, which did not instigate the change, to refresh the state presented to the user in their GUIs.

Table 3-259 Gesture configuration Changed notification

Notification ID	Version	ADK release	Variants
0	1	21.4	All

Table 3-260 Gesture configuration Changed notification parameters

Payload byte	Value	Comments							
1	Varies	There is a single-byte value specifying the 7-bit gesture ID that changes:							
		Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
		Reserved	Gesture ID for which the configuration was changed						

3.11.9 Configuration Reset To Defaults notification

The embedded application sends this message to all connected mobile applications when the UI configuration has been reset to its ROM default values. It has no payload.

Table 3-261 Configuration Reset to Defaults notification

Notification ID	Version	ADK release	Variants
1	1	21.4	All

Notification Parameters

None

3.12 Statistics reporting feature

This API has the notion of "Category IDs" and "Statistic IDs".

- A category ID refers to a grouping of related statistics - for example streaming statistics. Each grouping of statistics must have a unique category ID. To support the use of customer implemented extensions, category ID 0x8000 and above are reserved for customer use.
- A statistic ID identifies an individual statistic within a group, for example bitrate within the streaming statistic group.

To minimize BT traffic, when values are requested, more than one value may be requested within a single group.

Table 3-262 Statistics reporting feature

Feature ID	Version	ADK Release
0x0c	1	22.1

3.12.1 Get Supported Categories Command

This command is sent by the mobile application to discover the category IDs supported by the embedded device.

Table 3-263 Get Supported Categories command

Command ID	Version	ADK release	Variants
0	1+	22.1	All

Table 3-264 Get Supported Categories command parameters

Payload byte	Value	Comment
1 - 2	0 - 65535	Last received category ID. For first request this value is 0.

Table 3-265 Get Supported Categories command response PDU contents

Payload byte	Value	Comments
1	0 or 1	1 denotes there is more data to come after this response and a further request should be made passing the last received category id.
2 - 3	1 - 65535	Category ID. Category IDs are returned in numeric order, smallest first.
4 - 5	1 - 65535	Next category ID. List repeats until maximum allowed in response packet.

Possible Feature specific error status codes

None.

3.12.2 Get All Statistics for Category command

This command is sent by the mobile application to discover the statistic IDs supported by the embedded device.

Table 3-266 Get All Statistics For Category command

Command ID	Version	ADK release	Variants
1	1+	22.1	All

Table 3-267 Get All Statistics For Category command parameters

Payload byte	Value	Comments
1 - 2	1 - 65535	Category ID
3	0 - 255	Last received Statistic ID. For first request this value is 0.

Table 3-268 Get All Statistics For Category command response PDU contents

Payload byte	Value	Comments
1	0 or 1	1 denotes there is more data to come after this response and a further request should be made passing the last received statistic ID.
2 - 3	1 - 65535	Category ID
2 to max permitted in packet		List of statistics values in format described below (value entries do not include optional Category ID field for this command). Values are returned in numeric order of the statistic ID, smallest first.

Possible Feature specific error status codes

None

3.12.3 Get Statistics Values command

This command is sent by the mobile application to determine the values of a requested set of statistic IDs.

Table 3-269 Get Statistics Values command

Command ID	Version	ADK release	Variants
2	1+	22.1	All

Table 3-270 Get Statistics Values command parameters

Payload byte	Value	Comments
1 to max permitted in packet		List of statistic ID requests. The list shall consist of entries of the format: Bytes 1-2: Category ID (1 – 65535) Byte 3: Statistic ID (1 – 255)

Table 3-271 Get Statistics Values command response PDU contents

Payload byte	Value	Comments
1	0 or 1	Denotes whether the entire request was fulfilled. Unlike other requests this command does not support "continuation" requests. A new request passing a new list of command IDs should be made for the unfulfilled values. Values are returned in numeric order of the statistic ID, smallest first.
2 to max permitted in packet or required to fulfil the request.		List of statistics values in format described below

Table 3-272 Statistic value entry

Payload byte	Value	Comments
1 - 2	1 - 65535	Category ID (Optional, only present in responses to Get Statistics Values command)
3	1 - 255	Statistic ID
4	0 - 255	Flags (TBC but to include suitability for regular refresh)
5	0 - 255	Length of value (0 if not available)
6+		Value as described by the type above (1, 2 or 4 bytes). If the value is not available then there is no value and length is 0

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Document references

Document	Reference
<i>Using the Debug Partition Application Note</i>	80-CH505-1 / CS-00420649-AN

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hzq@itdtek.com

Terms and definitions

Term	Definition
ANC	Active Noise Cancellation
API	Application Programming Interface
Bluetooth SIG	Bluetooth Special Interest Group
DFU	Device Firmware Upgrade
DLE	Data Length Extension
GAIA	Generic Application Interface Architecture
GATT	Generic Attributes
iAP	iPod Accessory Protocol
LE	Low Energy
PDU	Protocol Data Unit
QTI	Qualcomm Technologies International, Ltd.
RFCOMM	Radio Frequency COMMunication
SOF	Start of Frame
VA	Voice Assistant